Toygar Tanyel

List of Publications

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My name in **bold**. † first author, * co-first author, ◇ led by a student I mentored.

Peer-Reviewed Conference & Workshop Papers

[1] ◇ P. M. Konrad, **T. Tanyel**, S. Ayvaz. "Routing Subspaces: Auditing Evaluation-to-Deployment Mismatch in Fine-Tuned Language Models." *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. main track

[2] P. M. Konrad, **T. Tanyel**, S. Ayvaz. "A Path Already Walked: On Inheriting Network-Neuroscience Tools for Mechanistic Interpretability." *ICML 2026 Workshop on Mechanistic Interpretability*. I conceived and led this work

[3] ◇ P. M. Konrad, **T. Tanyel**, S. Ayvaz. "Decoded but Unused: Instruction Tuning Routes Moral Framing into the Judgment Readout." *ICML 2026 Workshop on Mechanistic Interpretability*.

[4] ◇ P. M. Konrad, **T. Tanyel**, S. Ayvaz. "Self-Reports Do Not Identify Self-Models: An Identifiability Test for Counterfactual Reports." *ICML 2026 Workshop on Philosophy of Machine Learning (PhilML)*. oral

[5] † **T. Tanyel**, N. Denizoglu, M. E. Seker, D. Alis, E. Cerekci, E. Karaarslan, et al. "Mammographic Breast Positioning Assessment via Deep Learning." *Deep-Breast Workshop on AI and Imaging for Diagnostic and Treatment, MICCAI 2024*.

[6] † **T. Tanyel**, G. Yildiz, C. Aydinalp, I. Oksuz. "Estimation of Two-Dimensional Electric Field Distribution Through Deep Learning: Preliminary Study." *32nd IEEE Signal Processing and Communications Applications Conference (SIU)*, 2024.

[7] † **T. Tanyel**, B. Alkurdi, S. Ayvaz. "Linguistic-Based Data Augmentation Approach for Offensive Language Detection." *7th Int. Conf. on Computer Science and Engineering (UBMK)*, 2022.

[8] ◇ P. M. Konrad, **T. Tanyel**, S. Ayvaz. "Beyond Major Floods: Deep Learning for Detecting Shallow Water Inundation in Agricultural Areas." *Procedia Computer Science*, 2025.

[9] G. Babur Guler, Y. A. Er, A. Guler, H. Uysal, A. K. Kalkan, **T. Tanyel**, M. Erturk, et al. "Transfer Learning for Echocardiographic Detection of HFpEF: Preliminary Results of the TALE-HFpEF Study." *European Heart Journal – Cardiovascular Imaging*, 2025. abstract

Peer-Reviewed Journal Articles

[1] † **T. Tanyel**, S. Atmaca, K. Gokce, M. Y. Balik, A. Guler, E. Aslanger, I. Oksuz. "Interpretable ECG Analysis for Myocardial Infarction Detection through Counterfactuals." *Biomedical Signal Processing and Control*, 2025.

[2] N. Denizoglu*, **T. Tanyel***, E. C. Erkan, M. E. Seker, E. Meltem, et al. "Landmark Regression with Attention U-Net for Assessing Breast Positioning Quality in MLO Mammography View." *Diagnostics*, 2026.

[3] O. Camurdan*, **T. Tanyel***, E. Aktufan Cerekci, D. Alis, E. Meltem, N. Denizoglu, et al. "Annotation-Efficient, Patch-Based, Explainable Deep Learning Using a Curriculum Method for Breast Cancer Detection in Screening Mammography." *Insights into Imaging*, 2025.

[4] † **T. Tanyel**, C. Nadarajan, N. M. Duc, B. Keserci. "Deciphering Machine-Learning Decisions to Distinguish Between Posterior Fossa Tumor Types Using MRI Features." *Cancers*, 2023.

[5] E. Cerekci, D. Alis, N. Denizoglu, O. Camurdan, M. E. Seker, C. Ozer, ..., **T. Tanyel**, et al. "Quantitative Evaluation of Saliency-Based Explainable AI (XAI) Methods in Deep-Learning-Based Mammogram Analysis." *European Journal of Radiology*, 2024.

[6] D. Alis, **T. Tanyel**, E. Meltem, M. E. Seker, D. Seker, H. M. Karakas, et al. "Choosing the Right Artificial Intelligence Solutions for Your Radiology Department: Key Factors to Consider." *Diagnostic and Interventional Radiology*, 2024.

[7] † **T. Tanyel**, B. Alkurdi, S. Ayvaz. "Developing Linguistic Patterns to Mitigate Inherent Human Bias in Offensive Language Detection." *Turkish Journal of Electrical Engineering and Computer Sciences*, 2024.

Book Chapter

[1] † **T. Tanyel**, I. Oksuz. "Evaluation of Explainability Methods in Medical Image Analysis." In *Explainable Machine Intelligence in Healthcare*, 2025.

Preprints

[1] † **T. Tanyel**, S. Ayvaz. "Whose Effect Is It? How Objectives Inflate Learned Subspace Interventions." *arXiv:2611.05033*, 2026.

[2] † **T. Tanyel**, S. Ayvaz, B. Keserci. "Beyond Known Reality: Exploiting Counterfactual Explanations for Medical Research." *arXiv:2307.02131*, 2023. 27 citations